

Lesson 1

Dynamics of lumped parameter systems

1 Remarks on notation

In this course, large use will be made of matrix algebra, therefore a proper notation must be defined to represent matrices and vectors, i.e. matrices with one row or with one column. Scientific books often use bold characters to denote matrices and vectors. However, this cannot be rendered on the blackboard during lessons; furthermore, it is useful for didactic purposes to use a different notation to denote a matrix and a vector. Thus, the notation that will be used during lessons and in these notes is the following:

- A scalar quantity will be denoted by a letter (from the Arabic or Greek alphabet) without any additional sign. For instance, the potential energy of a mechanical system will be denoted by V .
- A column vector (a matrix with n rows and 1 column) will be denoted by a letter with an underscore. For instance, the vector collecting the independent coordinates of a mechanical system will be denoted by $\underline{x}$.
- A row vector (a matrix with 1 row and n columns) will normally be introduced as the transpose of a column vector, e.g. $\underline{x}^T$.
- A matrix with a generic number of rows and columns will be denoted by a letter into square brackets, e.g. the mass matrix of a mechanical system will be denoted by $[m]$.

2 Some useful concepts from the pre-requisites of the course

- **Kinematic variables:** this term will be used when referring to kinematic quantities that are used to define the position of the mechanical system, or other useful quantities like elongation of springs. A vector formed by kinematic variables will be denoted as $\underline{y}$; depending on the particular scope for which these variables are introduced, an appropriate subscript will be added to this symbol to specify the meaning of the variable.
- **Constraints:** a constraint is something that prevents the mechanical system from undergoing movements that could be possible if the constraint was not present. The presence of a constraint may be expressed by a set of (generally non-linear) equations relating the values of some kinematic variables and their time derivatives:

$$f(\underline{y}, \dot{\underline{y}}) = 0 \quad (1.1)$$

Besides the effect on the kinematics of the system, constraints have another important effect in that they generate additional forces, referred to as constraint forces. When studying the motion of a mechanical system, the constraint forces are typically unknowns.

- **Holonomic constraints:** for some types of constraints, the above equation may be “integrated”, in the sense that it may be expressed in terms of the kinematical variables $\underline{y}$ only, without involving their time derivatives. In this case, the constraint is called *holonomic* and the corresponding constraint equation takes the form:

$$f(\underline{y}) = 0 \quad (1.2)$$

From the point of view of the physical interpretation, a holonomic constraint is a constraint that prevents the system from assuming some configurations that would be instead possible if the constraint was not applied. Note the subtle difference between preventing the system from undergoing a specific type of motion (effect of a generic constraint, possibly non-holonomic) and preventing the system from assuming a given position (specific to holonomic constraint only).

Independent coordinates: if a mechanical system whose configuration is identified by n_y kinematical variables is subjected only to holonomic constraints, expressed by n_c constraint equations (obviously, n_c must be strictly lower than n_y , to allow the system to move) then it is possible to use the constraint equation in the form (1.2) to identify a sub-set of $n = n_y - n_c$ parameters $\underline{x}$ that may freely assume any value within a specified range without violating the constraints acting on the system (these will be called “independent coordinates” of the system), and to express all kinematical variables as functions of the independent coordinates in the form:

$$\underline{y} = \underline{f}(\underline{x})$$

Virtual displacement: starting from a given position of the mechanical system considered, a virtual displacement is a displacement characterised by the following three properties:

- i. it has infinitesimal amplitude;
- ii. it is compatible with the constraints acting on the system
- iii. it happens at frozen time (only important for time-varying constraints)

Principle of Virtual Work: this principle states that for a mechanical system in static equilibrium the total virtual work of all forces acting on the system is zero for all virtual displacements of the system.

3 Equations of motion for a mechanical system

Let us consider a mechanical system having n degrees of freedom. The equations of motion for this system are a set of n second-order differential equations in n unknowns represented by the system's independent coordinates $\underline{x}$. These equations describe the variation in time of the system's coordinates $\underline{x}$, i.e. the “motion” of the system.

Using a standard mathematical treatment, it is possible to transform the set of n second-order equations of motion into a state space form, i.e. into a set of $2n$ first-order differential equations.

Different methods can be followed to derive the equations of motion for a mechanical system, and this topic is included in the pre-requisites of the course. Nevertheless, in this section a short outline of the most commonly used methods is provided.

3.1 Approach 1: Newtonian dynamics

This approach is based on the well known “Newton's second law”, which, for a point mass, states:

$$\sum \vec{F} = m\vec{a} \quad (3.1)$$

with $\sum \vec{F}$ the sum of all forces applied on the point and $\vec{a}$ the acceleration of the point mass.

For one single rigid body in planar motion, Newton's second law is generalised to the following set of two vector equations:

$$\begin{cases} \sum \vec{F} = m\vec{a}_G \\ \sum \vec{M}_O = J_O \vec{\omega} \end{cases} \quad (3.2)$$

In the upper equation of (3.2) $\vec{a}_G$ is the acceleration of the centre of mass (c.o.m.) of the body. In the lower equation, $\vec{M}_O$ is the moment of all forces acting on the body referred to a point O called the “centre”, J_O is the moment of inertia of the body with respect to the same centre O and $\vec{\omega}$ is the angular velocity of the body.

Important note: the centre O in the second of equations (3.2) must be either a still point ($\vec{V}_O = 0$) or the centre of mass of the body.

For a body in planar motion both $\sum \vec{M}_O$ and $\vec{\omega}$ are vectors directed perpendicularly to the plane in which the movement of the body takes place.

For a system composed by one single rigid body, equations (3.2) provide 3 scalar equations, whereas the number n of the degrees of freedom can be up to 3, depending on the number and type of constraints applied, and the number of unknown force constraint components is always $3-n$. Therefore, the total number of unknowns in the problem is 3 and equals the number of available equations, but the system will normally need to be re-arranged to eliminate the constraint forces and derive as many equations of motion as needed in “pure” form, i.e. not involving unknown constraint forces.

Example:

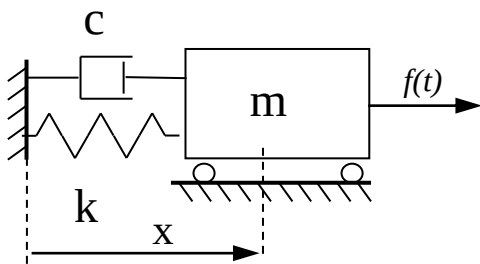


Figure 3.1: 1-dof system

Let us consider the 1-dof system in Figure 3.1. The forces acting on the mass m are represented in Figure 3.2 and the component along the horizontal direction of (3.1) brings:

$$f(t) - kx - c\dot{x} = m\ddot{x} \rightarrow \boxed{m\ddot{x} + c\dot{x} + kx = f(t)}$$

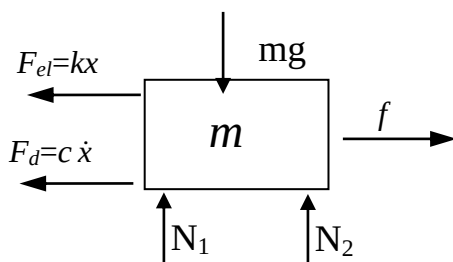


Figure 3.2: Scheme of forces acting on the system in Figure 3.1

3.2 Approach 1b: D'Alembert's principle

A different form of equations (3.2) consists of defining:

$$\begin{aligned} \vec{F}_{in} &= -m\vec{a} \\ \vec{M}_{in} &= -J_G \vec{\omega} \end{aligned} \quad (3.3)$$

with $\vec{F}_{in}$ defined as the inertia force and $\vec{M}_{in}$ defined as the moment of the inertia forces. Using (3.3), equations (3.2) are rewritten as:

$$\left\{ \begin{array}{l} \sum \vec{F} + \vec{F}_{in} = 0 \\ \sum \vec{M}_O + (G - O) \wedge \vec{F}_{in} + \vec{M}_{in} = 0 \end{array} \right. \quad (3.4)$$

The advantage of this form of the equations is that it makes apparent that any acceleration imposed to a mass will affect the balance of forces in the system. therefore, form (3.4) is often used instead of (3.2).

Example: Coming back to the example in Figure 3.1, Figure 3.3 shows the forces acting on the system, according to Equation (3.4).

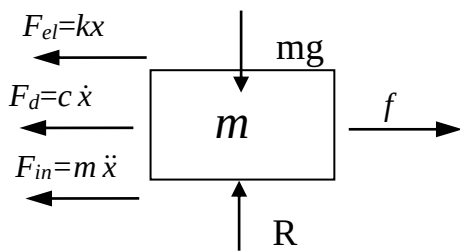


Figure 3.3: Scheme of forces acting on the system in Figure 3.1, according to Eq.s (3.3,3.4)

Taking the component along the horizontal direction of the vector equation we obtain:

$$f(t) - kx - c\dot{x} - m\ddot{x} = 0 \quad \rightarrow \quad \underline{m\ddot{x} + c\dot{x} + kx = f(t)}$$

3.3 Approach 2: the principle of virtual work

Using the D'Alembert's principle (3.4), the equations representing the movement of the mechanical system are written in the form of equilibrium equations. Hence, the principle of virtual work applies, provided that the work of the inertia force and inertia moment are taken into account:

$$\delta W + \delta W_{in} = 0 \quad \text{for any virtual displacement } \forall \delta \chi \quad (3.5)$$

It is possible to show that under the assumption of non-dissipative constraints (i.e. all constraint forces generate zero work, this is typical of frictionless constraints and of “rolling without sliding” constraints) Equation (3.5) brings a set of n independent equations of motion, with n the number of degrees of freedom, not including any additional unknown constraint force. For this reason, the approach based on the principle of virtual work is better suited than the approach based on Newton's second law (or D'Alembert's principle) to treat cases when several constraint forces are present in the system, see also Section 3.5.

Example: Once more we consider the example in Figure 3.1, with the forces shown in Figure 3.3 which include the inertia force F_{in} . The only possible virtual displacement compatible with the constraints applied on the sliding mass is a horizontal infinitesimal displacement δx , hence (3.5) brings:

$$f(t)\delta x - (kx + c\dot{x} + m\ddot{x})\delta x = 0$$

since the above equation must hold for any arbitrary value of the virtual displacement δx we obtain:

$$f(t) - kx - c\dot{x} - m\ddot{x} = 0 \quad \rightarrow \quad m\ddot{x} + c\dot{x} + kx = f(t)$$

3.4 Approach 2b: Lagrange equations

Lagrange equations represent a different form of the principle of virtual work, considering a special expression for the virtual work of some forces, namely conservative forces, dissipative forces, inertia forces.

As for the principle of virtual work, Lagrange equations are a set of n equations, as many as the number of the system's degrees of freedom. In case of non-dissipative constraints, the constraint forces do not appear in the equations of motions, that hence are the system's equations of motion.

Considering the case of 1-degree of freedom (d.o.f.) system, one single Lagrange equation can be written and takes the form:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) - \frac{\partial T}{\partial x} + \frac{\partial D}{\partial \dot{x}} + \frac{\partial V}{\partial x} = \frac{\delta W}{\delta x} \quad (3.6)$$

where x is the (single) independent coordinate adopted for the system, T is the kinetic energy, D the dissipation function, V the potential energy and W the virtual work of all forces other than elastic, viscous and inertial ones.

Lagrange equations for a multi-d.o.f. system are introduced in Section 6 of this Lesson.

Example: again we consider the example in Figure 3.1. For this system the kinetic energy, dissipation function, potential energy, virtual work and Lagrangian component are:

$$T = \frac{1}{2} m \dot{x}^2 ; D = \frac{1}{2} c \dot{x}^2 ; V = \frac{1}{2} k x^2 ; \delta W = f(t) \delta x$$

hence, applying (3.6) we obtain:

$$m\ddot{x} + c\dot{x} + kx = f(t)$$

3.5 Another example a system with “many” constraints

Based on the example in Figure 3.1, one could guess that there is no real advantage with using one or another of the methods available for writing the system's equations of motion. However, this is not the case and, as stated above, the use of the principle of virtual work or of Lagrange equations provides substantial benefits when the system addressed is composed by a relatively large number of bodies, interconnected by constraints. Furthermore, Lagrange equations provide a general procedure to write the equations of motion, that can be easily applied to a wide variety of systems, as will be shown later in this lesson.

For the time being, we underline the advantages of using Lagrange equations by considering the simple example of the slider-crank mechanism shown in Figure 3.4. For the sake of simplicity, we assume the connecting rod AB to be massless. We also assume all constraints to be frictionless.

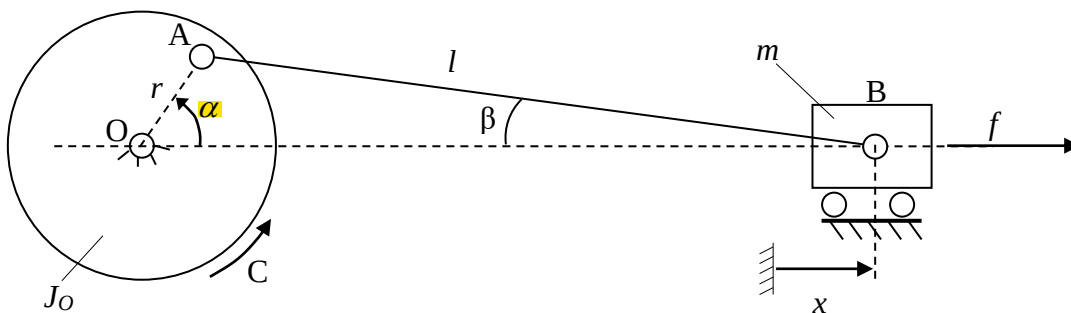


Figure 3.4: Slider-crank mechanism

We take the rotation of the crank α as the independent coordinate of the 1-dof system, and we express the position of the slider x as a kinematic variable function of the independent coordinate.

Considering the height of the triangle OAB, we get the following relationship:

$$r \sin(\alpha) = l \sin(\beta) \rightarrow \beta \cong \arcsin\left(\frac{r}{l} \sin(\alpha)\right)$$

We consider the case when $l \gg r$, so that the angle β is small and $\cos(\beta) \cong 1$. Based on this approximation, we get:

$$x \cong r \cos(\alpha) + l \cos(\beta) \cong r \cos(\alpha) + l \rightarrow \dot{x} \cong -r \sin(\alpha) \dot{\alpha} \rightarrow \ddot{x} \cong -r \sin(\alpha) \ddot{\alpha} - r \cos(\alpha) \dot{\alpha}^2$$

Using the approach based on Newtonian dynamics and D'Alembert's principle, one should consider the scheme of forces shown in Figure 3.5, which includes as unknowns not only the independent coordinate α and its derivatives, but also 8 constraints forces, denoted by $H_1, V_1, H_2, V_2, H_3, V_3, V_4, M_4$. To determine these quantities, one should write 3 equations in the form (3.4) for the crank, the connecting rod and the slider, with a total of 9 unknowns and 9 equations. This relatively long procedure is left to the reader.

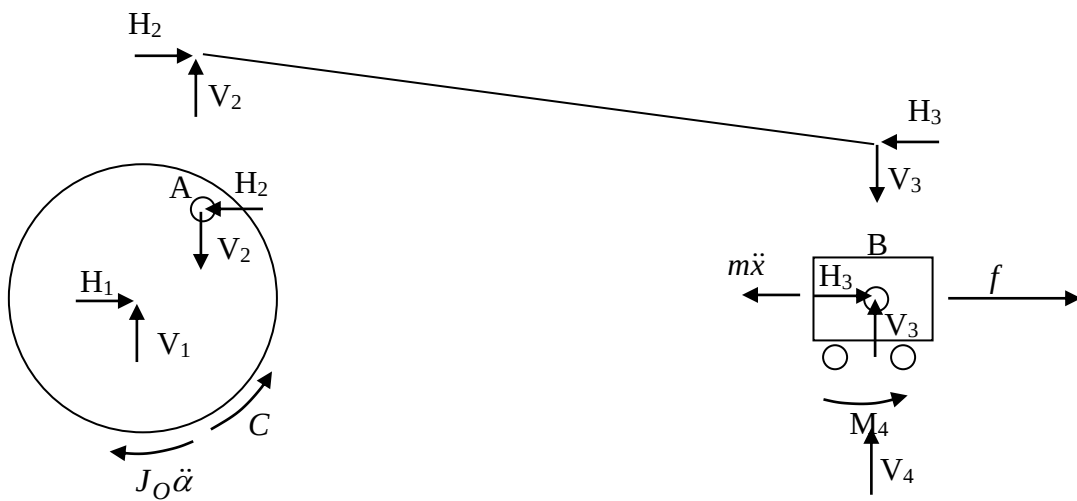


Figure 3.5: Scheme of forces for the slider-crank mechanism in figure 3.4 (massless connecting rod)

Using instead the method of Lagrange equations, the single equation of motion can be derived in a quite straightforward way. The kinetic energy for the system is:

$$T = \frac{1}{2} J_O \dot{\alpha}^2 + \frac{1}{2} m \dot{x}^2 = \frac{1}{2} (J_O + m r^2 \sin^2(\alpha)) \dot{\alpha}^2$$

The dissipation function and potential energy are both null functions, whereas the Lagrangian component of the moment C applied on the crank and of the force F applied on the slider is obtained from the virtual work as follows:

$$\delta W = C \delta \alpha + f \delta x = (C - f r \sin(\alpha)) \delta \alpha \rightarrow \frac{\delta W}{\delta \alpha} = C - f r \sin(\alpha)$$

Finally, applying Lagrange equation with respect to the independent coordinate α :

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\alpha}} \right) - \frac{\partial T}{\partial \alpha} + \frac{\partial D}{\partial \dot{\alpha}} + \frac{\partial V}{\partial \alpha} = \frac{\delta W}{\delta \alpha}$$

the equation of motion is obtained:

$$\left(J_O + m r^2 \sin^2(\alpha) \right) \ddot{\alpha} + m r^2 \sin(\alpha) \cos(\alpha) \dot{\alpha}^2 = C - f r \sin(\alpha)$$

4 Non-linear equations of motion for a system with 1 degree of freedom

Let us first consider a system with a single degree of freedom (d.o.f.), whose position is expressed by one single scalar coordinate x . In this section the complete (non-linear) equation of motion for this type of system is derived.

4.1 Inertial terms in the equations of motion

We treat first the inertial terms of the equation, and we start by demonstrating that for any 1 d.o.f. system the kinetic energy takes the form:

$$T = \frac{1}{2} m(x) \dot{x}^2$$

where $m(x)$ is called the *generalised mass* of the system and is, in general, a function of the independent coordinate x .

To this end, we consider a system formed by N_B rigid bodies moving in a plane. According to König's theorem, the kinetic energy of the system is:

$$T = \sum_{i=1, n_b} \frac{1}{2} (m_i \dot{x}_{G_i}^2 + m_i \dot{y}_{G_i}^2 + J_i \dot{\theta}_i^2) \quad ||$$

where $\dot{x}_{G_i}$ and $\dot{y}_{G_i}$ are the components of the speed of the centre of mass (c.o.m.) of the i -th body along the coordinate axes, $\dot{\theta}_i$ is the angular velocity of the body, M_i and J_i are respectively the mass and moment of inertia (referred to the c.o.m.) of the i -th body.

Since the system has 1 d.o.f., for each i -th body the position of the centre of mass G_i and the rotation θ_i of the body may be expressed as functions of one single independent coordinate x :

$$\begin{cases} x_{G_i} = x_{G_i}(x) \\ y_{G_i} = y_{G_i}(x) \\ \theta_i = \theta_i(x) \end{cases}$$

taking the time derivative of these relationships and using the chain rule brings:

$$\begin{cases} \dot{x}_{G_i} = \frac{dx_{G_i}}{dt} = \frac{\partial x_{G_i}}{\partial x} \frac{\partial x}{\partial t} = \lambda_{m_{i_x}}(x) \dot{x} \\ \dot{y}_{G_i} = \frac{dy_{G_i}}{dt} = \frac{\partial y_{G_i}}{\partial x} \frac{\partial x}{\partial t} = \lambda_{m_{i_y}}(x) \dot{x} \\ \dot{\theta}_i = \frac{d\theta_i}{dt} = \frac{\partial \theta_i}{\partial x} \frac{\partial x}{\partial t} = \lambda_{m_{i_\theta}}(x) \dot{x} \end{cases}$$

where $\lambda_{m_{i_x}}$, $\lambda_{m_{i_y}}$ and $\lambda_{m_{i_\theta}}$ are the derivatives with respect to coordinate x of the kinematical relationships expressing the c.o.m. position and body rotations, and are called “Jacobians” of the kinematical relationships.

By introducing the above equations in the expression of the kinetic energy, the following expression is obtained:

$$T = \sum_{i=1, n_b} \frac{1}{2} (m_i \dot{x}_{G_i}^2 + m_i \dot{y}_{G_i}^2 + J_i \dot{\theta}_i^2) = \frac{1}{2} \left[\sum_{i=1, n_b} (m_i \lambda_{m_{i_x}}^2(x) + m_i \lambda_{m_{i_y}}^2(x) + J_i \lambda_{m_{i_\theta}}^2(x)) \right] \dot{x}^2 = \frac{1}{2} m(x) \dot{x}^2 \quad (4.1)$$

which is the expected expression for the kinetic energy of the system. Furthermore, the above equation clarifies that the *generalised mass* of the system is obtained as the sum over the n_B bodies of the product of the inertial parameters of the body times the second power of the Jacobians. Since the Jacobians are in general a function of the coordinate x , also the generalised matrix will be a function of x .

As a very important exception to the above consideration, it may be observed that if all kinematical relationships expressing the c.o.m. position and rotation of all bodies are linear, then all the Jacobian functions will be constant, and thus the generalised mass of the system will not depend upon the coordinate x , but instead will be a constant parameter.

Taking then the derivatives of the kinetic energy prescribed by Lagrange equations, we obtain:



$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) - \frac{\partial T}{\partial x} = m(x) \ddot{x} + \frac{1}{2} \frac{dm}{dx} \dot{x}^2$$

4.2 Other terms in the equations of motion

We start recalling that the dissipation function D is defined as:

$$D = \sum_{i=1, n_d} \frac{1}{2} c_i (\dot{\Delta \ell}_i)^2 \quad (4.2)$$

with c_i the damping coefficient and $\dot{\Delta \ell}_i$ the speed of elongation of the i -th damper. Since the system has one d.o.f., the elongation of the i -th damper may be expressed as function of the single independent coordinate x by a proper kinematical relationship:

$$\Delta \ell_i = \Delta \ell_i(x)$$

and, by taking the time derivative:

$$\dot{\Delta \ell}_i = \frac{d\Delta \ell_i}{dt} = \frac{d\Delta \ell_i}{dx} \frac{\partial x}{\partial t} = \lambda_{ci}(x) \dot{x}$$

Bringing this expression in Eq. (4.2) gives:

$$D = \sum_{i=1, N_D} \frac{1}{2} c_i (\dot{\Delta \ell}_i)^2 = \frac{1}{2} \left[\sum_{i=1, N_D} c_i \lambda_{ci}^2(x) \right] \dot{x}^2 = \frac{1}{2} c(x) \dot{x}^2$$

and, by applying the derivative prescribed by the Lagrange equation:

$$\frac{\partial D}{\partial \dot{x}} = c(x) \dot{x}$$

As far as the terms associated with conservative forces are concerned, we start from the observation that for a 1 d.o.f. system the potential energy V will be a generic (normally non linear) function of the single independent coordinate x :

$$V = V(x)$$

when we will go into the details concerning the linearization of the system equation(s), it will be necessary to examine the different conservative forces that may be acting on the system (in particular, elastic and gravitational forces). For the time being however, it is sufficient to observe that the derivative of the potential energy $V(x)$ with respect to the independent coordinate x , prescribed by the Lagrange equation, will be a generic (often non-linear) function of x , that takes the physical meaning of the Lagrangian component of all conservative forces along the coordinate x , with the sign changed due to the fact that this term appears in the left hand side in equation (4.1):

$$\frac{\partial V}{\partial x} = -f_c(x)$$

Finally, the virtual work of the non-conservative and non dissipative forces acting on the system is computed. We assume here that non-conservative / non-dissipative forces acting on the system consist of a set of n_f time-dependent forces. The virtual work of these forces may be expressed as:

$$\delta W = \sum_{i=1, n_f} (\delta x_{fi} f_{ix}(t) + \delta y_{fi} f_{iy}(t))$$

where $f_{ix}(t)$ and $f_{iy}(t)$ are the Cartesian components of the i -th time dependent force applied on the system and δx_{fi} , δy_{fi} are the Cartesian components of the virtual displacement of the point at which the force is applied.

The position of the point at which the force is applied can be expressed as function of the single independent coordinate x :

$$\begin{cases} x_{fi} = x_{fi}(x) \\ y_{fi} = y_{fi}(x) \end{cases}$$

and, by taking the differentials of these relationships, the Cartesian components of the virtual displacement are obtained as:

$$\delta x_{fi} = \frac{\partial x_{fi}(x)}{\partial x} \delta x = \lambda_{fi}(x) \delta x$$

$$\delta y_{fi} = \frac{\partial y_{fi}(x)}{\partial x} \delta x = \lambda_{fiy}(x) \delta x$$

where $\lambda_{fi}(x)$ and $\lambda_{fiy}(x)$ are the **Jacobians of the kinematic relationships between the coordinates of the point and the single independent coordinate x** .

The virtual work may be then expressed as:

$$\delta W = \sum_{i=1, n_f} (\delta x_{fi} f_{ix}(t) + \delta y_{fi} f_{iy}(t)) = \sum_{j=1, n_f} (\underbrace{\lambda_{fi}(x) f_{ix}(t) + \lambda_{fiy}(x) f_{iy}(t)}_{f(x,t)}) \delta x = f(x,t) \delta x$$

4.3 **Final form** of the **equation of motion for a 1-d.o.f. system**

By collecting the expressions of the different terms in the Lagrange equation as obtained in the two paragraphs above, the single equation of motion of a 1-d.o.f. system is obtained in the form:

$$m(x) \ddot{x} + \frac{1}{2} \frac{dm}{dx} \dot{x}^2 + c(x) \dot{x} - f_c(x) = f(x,t) \quad (4.3)$$

where the first two terms at the left-hand side come from the derivatives of the kinetic energy in the Lagrange equation, and thus represent the inertial terms of the equation, the third term at the left hand represents the effect of dissipative forces, the last term at the right hand side represents the effect of conservative forces and the single term at the right-hand side of the equation represents the effect of time-dependent forces acting on the system.

Although this result has been obtained under the assumption of planar motion of the system, it can be shown that the single equation of motion of a 1 d.o.f. system performing a spatial motion also takes the form of Eq. (4.3).

5 **Linearization** of the equation of motion for a system with 1 degree of freedom

Equation (4.3) describes the **general motion of the 1 d.o.f. system** under the action of given forces and **prescribed initial conditions**:

$$\begin{cases} x|_{t=0} = x_{in} \\ \dot{x}|_{t=0} = \dot{x}_{in} \end{cases}$$

where $(x_{in}, \dot{x}_{in})$ are **given constants** representing the **initial position and velocity of the system** at time $t=0$, **when the motion starts**. The solution of equation (4.3) with the above initial conditions (the so-called “Cauchy problem” in the theory of differential equations) provides the “exact” motion of the system under study. However, **exception made for a very limited category of systems** (essentially, those that will be called hereafter **linear systems**), an **exact solution cannot be obtained by analytical means**. For instance, **even the equation of motion of a 1 d.o.f. free and un-damped pendulum**, having the form:

$$m\ell^2 \ddot{\theta} + mg\ell \sin(\theta) = 0$$

under general initial conditions, **cannot be found exactly**, and instead **it may only be approximated by means** e.g. of **perturbation techniques**, (see e.g. Meirovitch, Fundamentals of Vibration, chapter 11), at the cost of very complicated and cumbersome calculations. **When the complexity of the system increases** (in particular, when the system has more than 1 d.o.f.), the **use of these perturbation techniques or of other semi-analytical methods to treat non-linear equations of motion becomes almost unfeasible**, and the **only practical mean** to obtain an approximated solution for non-linear equations of motion is to **use numerical integration methods** (like the *Runge-Kutta* methods).

However, there are a number of disadvantages in using numerical integration to solve the equation(s) of motion of a mechanical system:

- a computer hardware and a specific solution algorithm will be needed;
- the accuracy of the solution is affected by the numerical method applied and the size of the time step adopted;
- in some cases, the numerical method may provide a divergent solution, i.e. a solution that deviates by a large amount from the correct one due to issues specific to the numerical method used;
- most important of all, the solution will be obtained in the form of a sequence of discrete values rather than as an analytic expression: this means for instance that it will be almost impossible to understand the effect on of a variation of one parameter (e.g. the mass of one body) on the movement of the system.

Thus, a numerical solution may be used to solve an *analysis problem* (that is, define how a known system will behave under known forcing conditions), but will be of very poor help to solve a *design problem* (that is, set the proper value to the parameters of the system to obtain the desired dynamic behaviour).

Fortunately, in most cases the system undergoes a motion of small amplitude around a steady state configuration (often, an equilibrium position). In this case, it is possible to assume that the difference between the steady state motion and the actual motion of the system is so small that using an expansion in series of power of the equation of motion, terms of order higher than one may be neglected. In this way, it is possible to approximate the equation of motion of the 1 d.o.f. system in the form of a linear, second-order differential equation (normally with constant parameters) that can be solved quite easily by analytical methods. Of course, the solution of the linearised equation will be only an approximation of the “exact” solution, but still will be representative of the actual motion of the system, provided of course that the assumption of small motion around the reference condition is respected.

In order to define the procedure used to perform the linearisation of the equation of motion Eq. (4.3), we first recall that an equilibrium position of the system is a configuration at which the system stands still, provided the system is initially placed at the equilibrium position with no initial velocity and it is not subjected to any time-dependent forces. It follows that all equilibrium positions are defined by a constant value of the system’s independent coordinate:

$$\begin{aligned} x &= x_0 = \text{const} \\ \dot{x} &= 0 \\ \ddot{x} &= 0 \end{aligned} \quad (4.4)$$

where the constant value x_0 satisfies a special form of the equation of motion (4.3) where the time dependent forces are set to zero and hence the right-hand side of the equation takes zero value (the so-called “homogeneous equation”). Bringing Eq. (4.4) in Eq. (4.3) and considering $f(x, t) = 0$ we obtain the following equation providing the equilibrium positions of the system:

$$f_c(x_0) = 0 \quad \text{EQUILIBRIUM} \quad (4.5)$$

This is an ordinary (non differential), non-linear equation which can be solved either by analytical or numerical means, depending on the system under study. Since Eq. (4.5) is non-linear, it is not possible to provide a general statement about the existence and uniqueness of the equilibrium position. Normally, mechanical systems do have at least one equilibrium position, but often they have multiple equilibrium positions. Note that, if the system has multiple equilibrium positions, the linearisation procedure described below for Eq. (4.3) leads to different results depending on the specific equilibrium position considered.

To perform the linearisation of Eq. (4.3) in the neighbourhood of a given equilibrium position represented by Eq. (4.4) we use the following procedure:

- i. we change the independent coordinate x to a new one $\bar{x} = x - x_0$ so that $\bar{x} = 0$ represents the position of equilibrium around which the equation is being linearised.
- ii. we expand each term of equation (4.3) in series of power with respect to variables $\bar{x}, \dot{\bar{x}}, \ddot{\bar{x}}$ in the neighbourhood of $\bar{x} = 0, \dot{\bar{x}} = 0, \ddot{\bar{x}} = 0$ and we truncate the series to first-order terms. Also the product of a time-dependent force $f(t)$ times the coordinate $\bar{x}$ in the right-hand side of equation (4.3) has to be neglected.

Though apparently simple, the above described linearization procedure is worth some attention and some remarks, and is described below for each term in equation (4.3). To keep the notation as simple as possible, we drop from now on the overscore for the system's independent coordinate x .

5.1 Linearization of the inertial terms

Let us first expand in series of power the generalised mass of the system $m(x)$:

$$m(x) \cong m(0) + \left. \frac{\partial m}{\partial x} \right|_0 x; \quad \frac{\partial m}{\partial x} \cong \left. \frac{\partial m}{\partial x} \right|_0$$

by substituting this approximation into the inertial terms of equation (4.1) and then neglecting all the residual terms of second order (i.e. containing $x\ddot{x}$ or $\dot{x}^2$), the following is obtained:

$$m(x)\ddot{x} + \frac{1}{2} \frac{dm}{dx} \dot{x}^2 \cong \left(m(0) + \cancel{\left. \frac{dm}{dx} \right|_0 x} \right) \ddot{x} + \frac{1}{2} \cancel{\frac{dm}{dx}} \dot{x}^2 \cong m(0)\ddot{x} \quad (4.6)$$

which demonstrates that the linearization of the inertial terms may be easily obtained by replacing the exact expression of the generalised mass as function of coordinate x with its value in the equilibrium position around which the equation is linearised.

This observation has an important consequence: since the dependence of the generalised mass from the coordinate x is due to the dependency of the Jacobian functions appearing in equation (4.3), the linearization of the inertial terms of the equations of motion may be obtained “a priori” by replacing the exact Jacobian functions appearing in the expression of the kinetic energy with their value in the equilibrium position, which means by using the linear approximation of the kinematical relationship relating the speeds of the c.o.m. and the angular speeds of the bodies to the time derivative of the independent coordinate $\dot{x}$. This conclusion is really important, because in many cases the linear approximation of this kinematical relationships may be found in a much easier way than their complete nonlinear expression. Examples will be given in practice sessions.

5.2 Linearization of the dissipative terms

In section 3 it was shown that the (generally nonlinear) dissipative term in the equation of motion of a 1 d.o.f. system is:

$$\frac{\partial D}{\partial \dot{x}} = c(x)\dot{x}$$

we now expand the function $c(x)$ in series of power of the coordinate x :

$$c(x) = c(0) + \left. \frac{\partial c}{\partial x} \right|_0 x + \dots$$

finally, we observe that in order to obtain a linearised equation of motion, the product $x\dot{x}$ between x and $\dot{x}$ shall be neglected, and thus:

$$c(x)\dot{x} = c(0)\dot{x} + \left. \frac{\partial c}{\partial x} \right|_0 x\dot{x} + \dots \cong c(0)\dot{x} \quad (4.7)$$

therefore, as with the inertial terms, a simple rule may be defined to linearise the dissipative terms in the equation of motion, that is to replace the function $c(x)$ expressing the dependence of the generalised damping from the coordinate x with the value of this function in the steady state position around which linearization is performed.

5.3 Linearization of the terms related to conservative forces

First of all, we observe that the linear approximation of the term $f_c(x)$ around the position $x=0$ is obviously:

$$f_c(x) \cong f_c(0) + \left. \frac{\partial f_c}{\partial x} \right|_0 x = f_c(0) - kx \quad \text{with} \quad k = - \left. \frac{\partial f_c}{\partial x} \right|_0$$

in the above expressions, the term k is called “generalised stiffness” of the system along the coordinate x . Thus, in principle, it could be possible to derive the exact expression of the Lagrangian component of all conservative forces and then obtain the generalised stiffness through the definition above.

However, there are at least two cases of conservative forces that deserve special attention and will be treated in this paragraph.

Elastic forces: elastic stiffness and second order elastic stiffness

Elastic forces are a particular case of conservative forces. In this course, only linear elasticity will be considered; moreover, in the first part of the course elastic deformation will be treated with reference to lumped elements (springs) whose inertia will be neglected. Thus we will confine in this lesson to the concept of “linear lumped spring”, e.g. a massless deformable body for which the force-deformation relationship is expressed by the linear expression:

$$F_{el} = -k_i \Delta \ell_i$$

where k_i is the stiffness of the spring (constant) and $\Delta \ell_i$ is the variation of length in the spring produced by the elastic deformation. Note $\Delta \ell_i$ is measured from the undeformed state of the spring, which may be different from the equilibrium position $x=0$; this means in general $\Delta \ell_i$ takes a non-zero value in the equilibrium position around which the linearisation is performed. Let us consider the simple case of the elastic force generated by a single i -th concentrated spring. The corresponding potential energy V_{el} has the form:

$$V_{el}(x) = \frac{1}{2} k_i \Delta \ell_i(x)^2$$

As for all other kinematical variables, $\Delta \ell_i$ may be expressed as function of the independent coordinate x .

In order to obtain a linearised expression for:

$$f_{c_{el}} = - \frac{\partial V_{el}(x)}{\partial x}$$

it is necessary to approximate the elastic potential energy to a quadratic form in x , which means expand V_{el} in series of power and neglect all terms of power greater than two. To this end, we first expand to the second order the function relating the dilatation of the spring $\Delta \ell_i$ with the independent coordinate x (we also recall that expansion must be performed around the zero value of the independent coordinate, which represents the equilibrium position of the system):

$$\Delta \ell_i(x) = \Delta \ell_i(0) + \left. \frac{\partial \Delta \ell_i}{\partial x} \right|_{x=0} x + \frac{1}{2} \left. \frac{\partial^2 \Delta \ell_i}{\partial x^2} \right|_{x=0} x^2 = \Delta \ell_{i0} + \lambda_{ki}(0)x + \frac{1}{2} \nu_{ki}(0)x^2$$

where λ_{ki} and ν_{ki} are respectively the Jacobian (first derivative) and Hessian (second derivative) of the kinematical relationship $\Delta \ell_i = \Delta \ell_i(x)$, evaluated at the equilibrium position $x=0$.

The second stage of the approximation is to bring the above expanded expression of the spring dilatation into the elastic potential energy, neglecting all terms proportional to x^3 or x^4 :

$$V_{el}(x) \cong \frac{1}{2} k_i \left(\Delta \ell_{i0} + \lambda_{ki}(0)x + \frac{1}{2} \nu_{ki}(0)x^2 \right)^2 \cong \frac{1}{2} k_i \Delta \ell_{i0}^2 + k_i \Delta \ell_{i0} \lambda_{ki}(0)x + \frac{1}{2} k_i \lambda_{ki}(0)^2 x^2 + \frac{1}{2} k_i \Delta \ell_{i0} \nu_{ki}(0)x^2$$

the above expression can be further simplified observing that:

- the first term is constant and will then disappear after the derivation prescribed by the Lagrange equation, it can then be neglected from the beginning;
- the second term (linear in x) must disappear due to the theorem of stationarity of the potential energy: this theorem states that a system subjected to conservative forces is in equilibrium in all and only those positions where the first derivative of the total potential energy is zero. Since we have set $x=0$ as our position of equilibrium, to have there a zero valued derivative of the potential energy we must require that the coefficient of the first order term in the power expansion of the total potential energy is zero. Of course, this does not necessarily mean that the linear term in V_{el} is zero, because the total potential energy will be the sum of different terms, due to different conservative forces (e.g. elastic and gravitational). However, when V_{el} will be summed with the terms due to the other conservative forces to form the total potential energy, the first order term in V_{el} will be necessarily cancelled by the first order term due to other forces (e.g. by an opposite first order term in the gravitational potential energy V_g).

by these simplifications, the expression of the elastic potential energy becomes:

$$\begin{aligned} V_{el}(x) &\cong \frac{1}{2} \left(k_i \lambda_{ki}^2(0) + k_i \Delta \ell_{i0} \nu_{ki}(0) \right) x^2 = \frac{1}{2} k_{el} x^2 \\ f_{c_{el}} &= -\frac{\partial V_{el}}{\partial x} = -k_{el} x \end{aligned} \quad (4.8)$$

where k_{el} represents the equivalent stiffness due to all elastic forces along the single coordinate x of the system.

Before concluding this section, it is important to spend some words on the form of the equivalent elastic stiffness k_{el} , which is in general the sum of two terms.

The first term $k_i \lambda_{ki}^2(0)$ depends on the first order approximation of the non-linear relationship $\Delta \ell_i(x)$ relating the elongation of the spring with the independent coordinate: in fact it contains the Jacobian λ_{ki} of this relationship. The second term $k_i \Delta \ell_{i0} \nu_{ki}(0)$ depends on the second order component of the non-linear relationship $\Delta \ell_i(x)$, as it contains the “Hessian” ν_{ki} . For this reason, it is sometimes referred as the “second order” elastic stiffness of the system. However, this terminology should not lead to the conclusion that this term will necessarily be small compared to the “first order” elastic stiffness $k_i \lambda_{ki}^2(0)$: there are cases where the effects of the two terms are comparable. For instance, in the suspensions of a railway vehicle, it is well known that the lateral stiffness of the suspension is higher in the tare condition than in the full load condition: this is due to the presence of a negative second order stiffness whose importance is greater for a greater load acting on the carbody, that produces a higher static deformation of the suspension $\Delta \ell_{i0}$.

It is important to observe that this second contribution to the equivalent elastic stiffness k_{el} disappears in all cases where at least one of the following conditions hold:

- the spring is unloaded (and thus undeformed) in the considered equilibrium position $x=0$, so that $\Delta \ell_{i0} = 0$;
- the Hessian of the kinematic relationship $\Delta \ell_i(x)$ is zero in the considered equilibrium position, so that $\nu_{ki}(0) = 0$. This second condition applies in particular to all cases where the relationship $\Delta \ell_i(x)$ is linear (in these cases the Hessian is zero for all values of x , not only in the steady state position).

Gravitational forces: the gravitational stiffness

To examine the effect of gravitational forces, we consider the case of one single i -th body subjected to the effect of gravity. The corresponding potential energy V_g has the form:

$$V_g(x) = m_i g h_i(x)$$

where m_i is the (gravitational) mass of the body, g is the gravitational acceleration ($g=9.81 \text{ m/s}^2$) and h_i is the “elevation” or vertical displacement of the body centre of mass above a given fixed reference.

In order to obtain a linearised expression for:

$$f_{c_g} = -\frac{\partial V_g(x)}{\partial x}$$

it is necessary to approximate the gravitational potential energy to a quadratic form in x and, since m_i and g are constant, this means to expand $h_i(x)$ to the second order:

$$h_i(x) = h_i(0) + \left. \frac{\partial h_i}{\partial x} \right|_{x=0} x + \frac{1}{2} \left. \frac{\partial^2 h_i}{\partial x^2} \right|_{x=0} x^2 = h_0 + \lambda_{gi}(0)x + \frac{1}{2} \nu_{gi}(0)x^2$$

where a notation of the same type as that introduced for the elastic terms has been used.

Bringing this expression into the one defining the gravitational potential energy, we get:

$$V_g(x) \cong m_i g h_0 + m_i g \lambda_{gi}(0)x + \frac{1}{2} m_i g \nu_{gi}(0)x^2$$

where again simplifications may be introduced by observing that:

- the first term is constant and will then disappear after the derivation prescribed by the Lagrange equation, it can then be neglected from the beginning;
- the second term (linear in x) must disappear due to the theorem of stationarity of the potential energy; this term will be indeed the one that cancels out with the linear term in the elastic potential energy.

by these simplifications, the expression of the elastic potential energy becomes:

$$V_g(x) \cong \frac{1}{2} m_i g \nu_{gi}(0)x^2 = \frac{1}{2} k_g x^2$$

$$f_{c_g} = -\frac{\partial V_g(x)}{\partial x} = -k_g x \quad (4.9)$$

where k_g represents the equivalent stiffness due to gravitational forces along the single coordinate x of the system, or simply the “gravitational stiffness”. Like for the second order elastic stiffness, the gravitational stiffness has non-zero values only in those cases where the Hessian of the kinematical relationship $h_i(x)$ has a non-zero value in the equilibrium position around which the equation of motion is linearised. Otherwise, no gravitational force will appear in the final linearised equation of motion. It can then be stated the following important conclusion:

If for a given system all kinematic relationships expressing the elevation of the c.o.m. of all bodies are linear in the independent coordinate(s), and if $x=0$ is chosen as the equilibrium position of the system, then gravitational forces will not appear in the equation(s) of motion of that system.

5.4 Linearization of the terms related to non-conservative forces

For the sake of simplicity, in the following expressions we consider the case of a single force $f_i(t)$; the general case may be easily obtained by extending (i.e., summing over the forces) the result below.

In section 4., we have shown that the Lagrangian component of a time-dependent force $f_i(t)$ acting on the system may be expressed as:

$$f(x, t) = \lambda_{fix}(x) f_{ix}(t) + \lambda_{fiy}(x) f_{iy}(t)$$

Additionally, we consider both Cartesian components of the force to have zero mean value (if this is not the case, each force component may be split into a zero-mean function of time and a constant force representing the non-zero mean value and this latter component may be treated as a conservative force).

Next, we expand Jacobians λ_{fix} and λ_{fiy} in series of power of the coordinate x :

$$\lambda_{fix} = \lambda_{fix}(0) + \left. \frac{\partial \lambda_{fix}}{\partial x} \right|_0 x + \dots$$

$$\lambda_{fiy} = \lambda_{fiy}(0) + \left. \frac{\partial \lambda_{fiy}}{\partial x} \right|_0 x + \dots$$

finally, in order to derive a linearised, constant coefficient equation of motion for the system, we state that both force components $f_{ix}(t)$, $f_{iy}(t)$ are ‘small’ so that the product of any of these quantities times any power of coordinate x is negligible with respect to the other linear terms of the equation. Thus, only the constant term in the above expansion must be retained, and the linearised expression of the time-dependent force is obtained as:

$$f(t) \cong \lambda_{fix}(0) f_{ix}(t) + \lambda_{fiy}(0) f_{iy}(t)$$

5.5 Final expression of the linearised equation of motion for a 1 d.o.f. system

By collecting the results of the above paragraphs, it is possible to state that the linearised form of the equation of motion will be in the form:

$$m_0 \ddot{x} + c_0 \dot{x} + kx = f_0(t)$$

where, recalling Eq. (4.6) to Eq. (4.9) the following notation is introduced:

$$m_0 = m(0) = \sum_{j=1, n_b} \left(m_i \lambda_{mi_x}^2(0) + m_i \lambda_{mi_y}^2(0) + J_i \lambda_{mi_\theta}^2(0) \right)$$

$$c_0 = c(0) = \sum_{i=1, n_d} c_i \lambda_{ci_x}^2(0)$$

$$k = k_{el_I} + k_{el_{II}} + k_g$$

$$f_0(t) = \sum_{i=1, n_f} \left(\lambda_{fix}(0) f_{ix}(t) + \lambda_{fiy}(0) f_{iy}(t) \right)$$

Furthermore, the three terms defining the total stiffness k are defined according to the following expressions (below, the general case of a system including n_s springs is considered):

$$k_{el_I} = \sum_{i=1, n_s} k_i \lambda_{ki}^2(0)$$

$$k_{el_{II}} = \sum_{i=1, n_s} k_i \Delta \ell_{i0} \nu_{ki}(0)$$

$$k_g = \sum_{j=1, n_b} m_j g \nu_{gi}(0)$$

5.6 Linear systems

We close this section by treating a particular class of system, where all kinematical relationships are linear by nature, which means that for a system of this type all the Jacobian functions defined in the paragraphs above are constant and all Hessian functions are identically zero. Owing to the above considerations, it is clear that a linear system will have the following properties:

1. the equation(s) of motion of the linear system will be linear, not only for small motion around a steady state position, but eventually also for large motion;
2. the “second order” elastic stiffness of any spring element will be zero, regardless the preload of the spring;
3. the gravitational stiffness will also be zero, and gravitational forces will not appear in the equation(s) of motion of the system, provided that the independent coordinate is chosen so that $x=0$ represents the equilibrium condition of the system¹;

¹ We observe by the way that a linear system will have one and only one equilibrium position. In fact, equilibrium configurations have to be stationary points for the total potential energy, but for a linear system the total potential energy will be exactly a quadratic function of the coordinate x , and thus will have one and only one single stationary point

6 Generalisation to systems with $n > 1$ degrees of freedom (multi-dof systems)

We next consider a multi-degree of freedom system, whose configuration has to be described by means of $n > 1$ independent coordinates, the considerations and conclusions already reported in sections 3 and 4 may be quite easily extended.

For a multi-degree of freedom system, Lagrange equations take the following form:

$$\left(\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_i} \right) - \frac{\partial T}{\partial x_i} + \frac{\partial D}{\partial \dot{x}_i} + \frac{\partial V}{\partial x_i} = f_i \right) \quad (6.1)$$

where:

- x_i is the i -th independent coordinate of the system, with $i=1, 2, \dots, n$;
- T is the kinetic energy of the system;
- D is the “dissipation function” of the system, accounting for dissipative forces produced in the system by viscous dampers;
- V is the potential energy associated to conservative forces acting on the system
- f_i is the Lagrangian component along the i -th coordinate of all forces different from conservative and dissipative ones: in our case, time-dependent forces.

It is recalled that, if δW is the virtual work of a force, the Lagrangian components of the force along all the independent coordinates of the system are defined by the relationship:

$$\delta W = \sum_{i=1, n} \delta x_i f_i$$

that is, the Lagrangian component along the i -th coordinate is obtained as the ratio between the virtual work produced by the force for a virtual displacement of the system that corresponds to a virtual variation δx_i of the i -th coordinate alone, divided by the amplitude of the virtual variation δx_i .

When dealing with a multi-d.o.f. system, it is advisable to make use of matrix notation. To do so, we first need to introduce in section 6.1 the definition of derivative with respect to a vector. Based on this definition it will be possible then to derive the complete non-linear and linearised equations of motion for a multi-d.o.f. system.

6.1 Derivation with respect to a vector

We define the derivative of a mathematical entity with respect to a vector $\underline{x}$ as the product of that generic mathematical entity times a row vector formed by the partial derivatives with respect to the elements $x_1, x_2, \dots$ of $\underline{x}$:

$$\frac{\partial \bullet}{\partial \underline{x}} = \bullet \left\{ \frac{\partial}{\partial x_1} \quad \frac{\partial}{\partial x_2} \quad \dots \quad \frac{\partial}{\partial x_n} \right\}$$

let us explain this definition by some examples:

Derivative of a scalar w.r.t. a vector: let us consider a scalar V , function of $x_1, x_2, \dots, x_n$, this can be denoted by: $V=V(\underline{x})$, then the derivative of V with respect to $\underline{x}$ takes the form:

$$\left\| \frac{\partial V}{\partial \underline{x}} = \left\{ \frac{\partial V}{\partial x_1} \quad \frac{\partial V}{\partial x_2} \quad \dots \quad \frac{\partial V}{\partial x_n} \right\} \right\| \quad (6.2)$$

this definition brings immediately an important advantage: the “scalar” form of the Lagrange equations for a system with n d.o.f. is:

$$\begin{cases} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_1} \right) - \frac{\partial T}{\partial x_1} + \frac{\partial D}{\partial \dot{x}_1} + \frac{\partial V}{\partial x_1} = f_1 \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_2} \right) - \frac{\partial T}{\partial x_2} + \frac{\partial D}{\partial \dot{x}_2} + \frac{\partial V}{\partial x_2} = f_2 \\ \dots \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}_n} \right) - \frac{\partial T}{\partial x_n} + \frac{\partial D}{\partial \dot{x}_n} + \frac{\partial V}{\partial x_n} = f_n \end{cases}$$

now, with the above definition of the derivative of a scalar w.r.t. a vector we observe that the n scalar equations above may be expressed by a single matrix equation, that takes the name of “**matrix form of the Lagrange equations**”:

$$\left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\underline{x}}} \right)^T - \left(\frac{\partial T}{\partial \underline{x}} \right)^T + \left(\frac{\partial D}{\partial \dot{\underline{x}}} \right)^T + \left(\frac{\partial V}{\partial \underline{x}} \right)^T = \underline{f} \right]$$

where $\underline{f}$ is the vector containing the Lagrangian components of time dependent forces along all the coordinates of the system, and is defined by the matrix expression:

$$\delta W = (\delta \underline{x})^T \underline{f}$$

Derivative of a vector w.r.t. a vector: let us consider a vector $\underline{y}$, whose elements $y_1, y_2, \dots, y_m$ are function of the elements $x_1, x_2, \dots, x_n$ of another vector $\underline{x}$ this can be denoted by: $\underline{y} = \underline{y}(\underline{x})$, then the derivative of $\underline{y}$ with respect to $\underline{x}$ is an m by n matrix in the form:

$$\frac{\partial \underline{y}}{\partial \underline{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \dots & \frac{\partial y_1}{\partial x_n} \\ \dots & \dots & \dots & \dots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \dots & \frac{\partial y_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial y_1}{\partial \underline{x}} \\ \vdots \\ \frac{\partial y_m}{\partial \underline{x}} \end{bmatrix}$$

this matrix is usually called the “**Jacobian matrix**” (or simply Jacobian) of the relationship between $\underline{y}$ and $\underline{x}$.

A remarkable result: we conclude this section by applying the above definitions to the calculation of the following derivative:

$$\frac{\partial (\underline{z}(\underline{x})^T [A] \underline{y}(\underline{x}))}{\partial \underline{x}}$$

where $[A]$ is a constant-valued, square and symmetric matrix of order n , while $\underline{y}$ and $\underline{z}$ are two column vectors of length n . As with the derivative of scalar quantities, the derivative of a product is the sum of two terms, one containing the derivative of $\underline{y}$ w.r.t. to $\underline{x}$, the second containing the derivative of $\underline{z}$ w.r.t. to $\underline{x}$. The first of these two terms can be easily obtained, since $\underline{y}$ is at the right hand side of the triple product, and thus may be multiplied by the differential operator defined in formula (5.1). In order to obtain the other derivative, one must observe that the triple product $(\underline{z}(\underline{x}))^T [A] \underline{y}(\underline{x})$ is a scalar quantity (i.e. a 1x1 matrix) and thus it coincides with its transpose:

$$(\underline{z}(\underline{x}))^T [A] \underline{y}(\underline{x}) = ((\underline{z}(\underline{x}))^T [A] \underline{y}(\underline{x}))^T = (\underline{y}(\underline{x}))^T [A]^T \underline{z}(\underline{x})$$

in this way, the $\underline{z}$ vector is transferred at the right hand side of the product and the second term of the derivative may be computed:

$$\frac{\partial (\underline{z}(\underline{x})^T [A] \underline{y}(\underline{x}))}{\partial \underline{x}} = (\underline{z}(\underline{x}))^T [A] \frac{\partial \underline{y}}{\partial \underline{x}} + (\underline{y}(\underline{x}))^T [A]^T \frac{\partial \underline{z}}{\partial \underline{x}}$$

If we consider the special case $\underline{y} = \underline{z} = \underline{x}$, note that, by definition, the derivative of the vector $\underline{x}$ with respect to the same vector is an identity matrix of order n , and recall that $[A]^T = [A]$, we get:

$$\frac{\partial (\underline{x}^T [A] \underline{x})}{\partial \underline{x}} = \underline{x}^T [A] [I] + \underline{x}^T [A]^T [I] = 2 \underline{x}^T [A]$$

6.2 Systematic approach to writing the equations of motion for a n -d.o.f. system

By the use of Lagrange equations and of matrix formalism, it is possible to define a systematic approach to writing the equations of motion of a multi-d.o.f. system, in the sense that these equations may be obtained by a fixed sequence of operations, regardless to the type of system considered (number of bodies, number and type of constraints, number of degrees of freedom). In the case that linearised equations of motion are sought (or in the case of a linear system), the result of this systematic procedure is a system of linear, second-order differential equations with the form (6.3).

Kinetic energy and derivation of the inertial terms in the equations of motion

For a general, multi-d.o.f. system formed by rigid bodies moving in a common plane, the kinetic energy will have the form:

$$T = \sum_{i=1, n_b} \frac{1}{2} (m_i \dot{x}_{G_i}^2 + m_i \dot{y}_{G_i}^2 + J_i \dot{\theta}_i^2)$$

we may rewrite this expression using a matrix notation by defining a vector of kinematical variables $\dot{\underline{y}}_m$ and a matrix of inertial parameters $[m_{ph}]$ with the following expressions:

$$\dot{\underline{y}}_m = \begin{Bmatrix} \dot{x}_{G_1} \\ \dot{y}_{G_1} \\ \dot{\theta}_1 \\ \vdots \\ \dot{\theta}_{n_b} \end{Bmatrix} ; \quad [m_{ph}] = \text{diag} \begin{bmatrix} m_1 & m_1 & J_1 & \dots & J_{n_b} \end{bmatrix}$$

it is then easy to recognise that, by the usual rule of matrix products, the above “scalar” expression of the kinetic energy (based on a sum over the bodies) may be rewritten as the following matrix product:

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [m_{ph}] \dot{\underline{y}}_m$$

then, we observe that the kinematical variables contained in vector $\dot{\underline{y}}_m$ are the time derivatives of the coordinates and angles of rotation of all bodies in the system, and that these quantities may be expressed by proper kinematical relationships (case by case, linear or non-linear):

$$\underline{y}_m = \begin{Bmatrix} x_{G_1} \\ y_{G_1} \\ \theta_1 \\ \vdots \\ \theta_{n_b} \end{Bmatrix}$$

and thus, by taking the time derivative of this vector (and remembering the definition of the Jacobian matrix as the derivative of vector $\underline{y}_m$ w.r.t. vector $\underline{x}$):

$$\dot{\underline{y}}_m = \frac{d\underline{y}_m}{dt} = \frac{\partial \underline{y}_m}{\partial \underline{x}} \frac{d\underline{x}}{dt} = [\lambda_m(\underline{x})] \dot{\underline{x}}$$

where $[\lambda_m(\underline{x})]$ is the Jacobian matrix of the kinematical relationships $\underline{y}_m = \underline{y}_m(\underline{x})$.

Bringing the above result in the matrix expression of the kinetic energy, we get:

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [m_{ph}] \dot{\underline{y}}_m = \frac{1}{2} \dot{\underline{x}}^T [\lambda_m(\underline{x})]^T [m_{ph}] [\lambda_m(\underline{x})] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [m(\underline{x})] \dot{\underline{x}}$$

If we want to linearise the equations of motion, as demonstrated in section 5 for the single d.o.f. system, we need to approximate the configuration-dependent mass matrix $[m(\underline{x})]$ to a constant matrix obtained by substituting $\underline{x}=\underline{x}_0$, that is the constant value of the system coordinates in the steady state configuration around which linearization is performed (we observe that if the system is linear, the mass matrix will be constant, and no approximation will be needed).

Below, we assume the equilibrium position around which the equations are linearised is $\underline{x}_0=\underline{0}$. If this is not the case, a change of the vector of variables can be performed, as shown for the single d.o.f. system in Section 5.

Hence, we introduce the following approximation for the kinetic energy of the system:

$$T \cong \frac{1}{2} \dot{\underline{x}}^T [\lambda_m(\underline{0})]^T [m_{ph}] [\lambda_m(\underline{0})] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [m(\underline{0})] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [m_0] \dot{\underline{x}}$$

and, by applying the Lagrange equations in matrix form (and recalling the results obtained in section 5.1):

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\underline{x}}} \right)^T - \left(\frac{\partial T}{\partial \underline{x}} \right)^T = [m_0] \ddot{\underline{x}}$$

It is important to remark that, in case the linearised equations are sought, it is not necessary to derive the Jacobian matrix as a function of the independent coordinates $\underline{x}$ and instead this matrix can be directly approximated by its value in the steady state configuration considered for the linearization. Since steady state / equilibrium positions are notable configurations of the system (e.g., for a simple pendulum, equilibrium positions will correspond to a vertical alignment of the pendulum) it may be much easier to define the Jacobian directly in the equilibrium position, rather than to derive a general expression for this matrix.

Dissipation function

Assuming the system includes a generic number N_d of linear viscous dampers, the dissipation function is:

$$D = \sum_{i=1, n_d} \frac{1}{2} c_i \Delta \dot{\ell}_i^2$$

we rewrite this expression in matrix form by introducing a vector of kinematical variables $\underline{y}_D$ and a diagonal matrix $[c_{ph}]$ formed by the viscous damping coefficients of the n_d dampers:

$$\underline{y}_D = \begin{Bmatrix} \Delta \ell_1 \\ \vdots \\ \Delta \ell_{n_d} \end{Bmatrix} ; \quad [c_{ph}] = \text{diag} [c_1 \quad \dots \quad c_{n_d}]$$

so that the expression of D becomes:

$$D = \frac{1}{2} \dot{\underline{y}}_D^T [c_{ph}] \dot{\underline{y}}_D$$

Next, we have

$$\dot{\underline{y}}_D = \frac{d\underline{y}_D}{dt} = \frac{\partial \underline{y}_D}{\partial \underline{x}} \frac{d\underline{x}}{dt} = [\lambda_d(\underline{x})] \dot{\underline{x}}$$

where $[\lambda_d]$ is the Jacobian matrix of the kinematical relationships defining the elongations of the dampers. Bringing the above in the expression of dissipation function, we obtain:

$$D = \frac{1}{2} \dot{\underline{x}}^T [c_{ph}] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [\lambda_d(\underline{x})]^T [c_{ph}] [\lambda_d(\underline{x})] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [c(\underline{x})] \dot{\underline{x}}$$

In the same way as seen for a 1-d.o.f. system, to linearise the equations of motion we simply replace the general expression of the configuration-dependent damping matrix $[c(\underline{x})]$ with its value in the equilibrium position $\underline{x}=0$:

$$D \cong \frac{1}{2} \dot{\underline{x}}^T [\lambda_d(0)]^T [c_{ph}] [\lambda_d(0)] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [c(0)] \dot{\underline{x}} = \frac{1}{2} \dot{\underline{x}}^T [c_0] \dot{\underline{x}}$$

and, by applying the Lagrange equations in matrix form:

$$\left(\frac{\partial D}{\partial \dot{\underline{x}}} \right)^T = [c_0] \dot{\underline{x}}$$

Elastic potential energy and derivation of the elastic part of the stiffness matrix

If the system includes a generic number n_s of springs, the elastic potential energy will be:

$$V_{el} = \sum_{i=1, n_s} \frac{1}{2} k_i \Delta \ell_i^2$$

if the system is non-linear, and the complete equations of motions are sought, then the use of a matrix formalism will be of very little help (or even more cumbersome). However, if the system is linear or if the linearised equations are sought, then it is possible to follow a matrix approach. To this end, as with the kinetic energy, we rewrite the scalar expression of the elastic potential energy by defining a vector of kinematical variables $\underline{y}_k$ and a diagonal matrix $[k_{ph}]$ formed by the stiffness parameters of n_s of springs in the system:

$$\underline{y}_k = \begin{Bmatrix} \Delta \ell_1 \\ \vdots \\ \Delta \ell_{n_s} \end{Bmatrix} ; \quad [k_{ph}] = \text{diag} [k_1 \quad \dots \quad k_{n_s}]$$

and thus:

$$V_{el} = \frac{1}{2} \underline{y}_k^T [k_{ph}] \underline{y}_k$$

now, from this matrix expression, it is easy to obtain the “first order” component of the elastic stiffness matrix, which depends (see section 4.2) upon the Jacobian of the relationships relating the elongations of the springs $\underline{y}_k$ with the independent coordinates $\underline{x}$. To this aim, we linearise the relationship $\underline{y}_k = \underline{y}_k(\underline{x})$ around the considered equilibrium position:

$$\underline{y}_k(\underline{x}) \cong \underline{y}_k(0) + \left. \frac{\partial \underline{y}_k}{\partial \underline{x}} \right|_0 \underline{x} = \underline{\Delta \ell}_0 + [\lambda_k(0)] \underline{x}$$

and, by substituting in the matrix expression of the potential energy:

$$V_{el} = \frac{1}{2} \underline{y}_k^T [k_{ph}] \underline{y}_k \cong \frac{1}{2} \underline{\Delta \ell}_0^T [k_{ph}] \underline{\Delta \ell}_0 + \underline{\Delta \ell}_0^T [k_{ph}] [\lambda_k(0)] \underline{x} + \frac{1}{2} \underline{x}^T [\lambda_k(0)]^T [k_{ph}] [\lambda_k(0)] \underline{x}$$

considering the above result, it is clear that the first term is constant and may thus be neglected, the second term is linear in the independent coordinates and will then disappear when summed into the total potential energy (provided as usual that the coordinates are chosen so that $\underline{x}=0$ is the equilibrium position), and the only term to be retained is the third one that may be rewritten as:

$$V_{el} \cong \frac{1}{2} \underline{x}^T [\lambda_k(0)]^T [k_{ph}] [\lambda_k(0)] \underline{x} = \frac{1}{2} \underline{x}^T [k_{ell}] \underline{x} \quad \text{with} \quad [k_{ell}] = [\lambda_k(0)]^T [k_{ph}] [\lambda_k(0)]$$

and by applying the Lagrange equations:

$$\left(\frac{\partial V_{el}}{\partial \underline{x}} \right)^T = [k_{ell}] \underline{x}$$

however, one should be very careful about the above result because, as demonstrated by the case of the 1-d.o.f. system, the matrix $[k_{ell}]$ obtained by the above procedure does not represent the whole

elastic stiffness matrix, but only the “first order” component. A second order component has to be summed in all cases where at least one spring in the system has a non-zero static load in the equilibrium position and the kinematical relationship expressing its elongation as function of the independent coordinates $\underline{x}$ has a non-zero Hessian matrix in the equilibrium position.

Unfortunately, since the calculation of the Hessian matrix involves the evaluation of the second derivatives of the kinematical relationship, the use of a matrix notation to generalise the case of the 1-d.o.f. system is not straightforward. However, it may be demonstrated that the second order component of the elastic stiffness matrix assumes the form:

$$\llbracket [k_{ell}] = \sum_{i=1, n_s} \Delta \ell_i(0) k_i [\nu_{ki}(0)] \rrbracket$$

where $[\nu_{ki}(0)]$ is the Hessian matrix ($n \times n$) containing the second derivatives of the relationship $\Delta \ell_i = \Delta \ell_i(\underline{x})$. Examples of the procedure to define the second-order component of the elastic stiffness matrix will be given in the practices.

Gravitational potential energy and derivation of the gravitational part of the stiffness matrix

If the system includes a generic number n_b of bodies (normally, all subjected to gravitational forces), the gravitational potential energy is:

$$\llbracket V_g = \sum_{i=1, n_b} m_i g h_i(\underline{x}) \rrbracket$$

if the system is non-linear, and the complete equations of motions are sought, then the use of a matrix formalism turns out to be almost useless. If instead the linearised equations are sought, then it is possible to treat this term as the second order component of the elastic potential energy and thus to derive a second order approximation of the gravitational potential energy in the form:

$$V_g \cong \frac{1}{2} \underline{x}^T [k_g] \underline{x}$$

with:

$$[k_g] = \sum_{i=1, n_b} m_i g [\nu_{hi}(0)]$$

where $[\nu_{hi}(0)]$ is the Hessian matrix ($n \times n$) containing the second derivatives of the relationship $h_i = h_i(\underline{x})$. Examples of the procedure to be followed in order to define the second-order component of the elastic stiffness matrix will be given in practice sessions.

By the way, it is observed that for a linear system, where the relationships $h_i = h_i(\underline{x})$ are linear for all the n_b bodies, the gravitational stiffness matrix is zero.

Lagrangian components of the time-dependent forces acting on the system

Considering the case where N_F time dependent forces are acting on the system, the corresponding virtual work will be:

$$\delta W = \sum_{j=1, n_f} (\delta x_{fj} f_{ix}(t) + \delta y_{fj} f_{iy}(t))$$

as usual, we rewrite this sum in matrix form by defining:

$$\delta \underline{x}_f = \begin{Bmatrix} \delta x_{f1} \\ \delta y_{f1} \\ \vdots \\ \delta x_{f_{n_f}} \\ \delta y_{f_{n_f}} \end{Bmatrix} \quad \text{and} \quad \underline{y}_f(t) = \begin{Bmatrix} f_{1x}(t) \\ f_{1y}(t) \\ \vdots \\ f_{n_fx}(t) \\ f_{n_fy}(t) \end{Bmatrix}$$

and thus:

$$\delta W = \delta \underline{x}_f^T \underline{y}_f(t)$$

then, we take the differential of the kinematical relationship $\underline{x}_f = \underline{x}_f(\underline{x})$:

$$\delta \underline{x}_f = \frac{\partial \underline{x}_f}{\partial \underline{x}} \delta \underline{x} = [\lambda_f(\underline{x})] \delta \underline{x}$$

where $[\lambda_f(\underline{x})]$ is a n_f by n rectangular matrix representing the Jacobian of the kinematical relationship $\underline{x}_f = \underline{x}_f(\underline{x})$. If we want then to obtain linear, time independent equations of motion, we approximate this relationship replacing the complete approximation of the Jacobian matrix by its value in the equilibrium position around which linearization is performed:

$$\delta \underline{x}_f \cong [\lambda_f(\underline{0})] \delta \underline{x}$$

bringing this result into the matrix expression of the virtual work, we get:

$$\delta W = \delta \underline{x}_f^T \underline{y}_f(t) \cong \delta \underline{x}^T [\lambda_f(\underline{0})]^T \underline{y}_f(t) = \delta \underline{x}^T \underline{f}_0(t)$$

with:

$$\underline{f}_0(t) = [\lambda_f(\underline{0})]^T \underline{y}_f(t)$$

thus, the Lagrangian components of the time-dependent forces acting on the system may be easily obtained by evaluating in the equilibrium position the Jacobian relating the displacements of the points of application of the forces with the independent coordinates and then performing the matrix product reported above.

Expression of the linearised equations of motion for a multi-d.o.f. system

Considering any system having n degrees of freedom (with $\underline{x}$ the vector of n independent coordinates) subject to inertial, linear viscous, linear elastic, gravitational and time-dependent forces, based on the results obtained in the previous paragraphs the kinetic energy T , dissipation function D , potential energy V and virtual work δW are approximated by the following expressions in order to obtain the linearised equation of motion of the system:

$$T = \frac{1}{2} \dot{\underline{x}}^T [m_0] \dot{\underline{x}} ; \quad D = \frac{1}{2} \dot{\underline{x}}^T [c_0] \dot{\underline{x}} ; \quad V = \frac{1}{2} \underline{x}^T [k] \underline{x} ; \quad \delta W = \delta \underline{x}^T \underline{f}_0(t)$$

considering the above result and applying the matrix form of the Lagrange equations it is easy to observe that:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\underline{x}}} \right)^T - \left(\frac{\partial T}{\partial \underline{x}} \right)^T = [m_0] \ddot{\underline{x}} ; \quad \left(\frac{\partial D}{\partial \dot{\underline{x}}} \right)^T = [c_0] \dot{\underline{x}} ; \quad \left(\frac{\partial V}{\partial \underline{x}} \right)^T = [k] \underline{x}$$

and thus that the Lagrange equations for the considered system will take the matrix form:

$$\| [m_0] \ddot{\underline{x}} + [c_0] \dot{\underline{x}} + [k] \underline{x} = \underline{f}_0(t) \| \quad (6.3)$$

where it is important to point out that this result applies regardless to the number of degrees of freedom of the system and that writing the set of equations of motion for the system is based on defining appropriate Jacobian matrices (and Hessian matrices, if required) and performing some matrix products involving these Jacobian / hessian matrices and vectors and matrices defined based on the properties of the system and on the forces acting on the system. We then envisage the possibility to use a matrix formalism to write the (linearised) equations of a n -d.o.f. system in a semi-automated way, thus minimising the possibility of errors and even with the possibility to implement the procedure in a computer code.